

Vantage Alpin: Data Generating Process Specification

Julian Hilgemann

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Goal. Simulate a realistic alpine e-commerce environment with $N_c = 5,000$ customers, $N_p = 500$ SKUs across three shops (DE, AT, CH) over a $T = 731$ -day horizon. The process couples heterogeneous customer activity, category-specific pricing, multiplicative seasonality, and a non-homogeneous Poisson arrival model to produce transactional fact tables suitable for dimensional modelling.

I. Customer base

Each customer c_i ($i = 1, \dots, N_c$) receives an activity score drawn from a Zipfian (power-law) distribution:

$$A_i \sim \text{Zipf}(a = 2.0).$$

This encodes a long-tail frequency structure: a small cohort drives the majority of transactions. The probability that customer c_i is selected for any given order is

$$P(c_i) = \frac{A_i}{\sum_{j=1}^{N_c} A_j}.$$

II. Product catalog & pricing

Products are allocated across three categories: *Bekleidung* (40%), *Schuhe* (30%), *Ausrüstung* (30%).

Base prices. Prices are drawn from category-specific log-normal distributions, preventing negative values and producing a realistic right-skewed tail:

Category	Distribution	Clip interval	Return prob. r
Schuhe	Lognormal($\mu=4.8$, $\sigma=0.5$)	[20, 800]	0.30
Bekleidung	Lognormal($\mu=4.4$, $\sigma=0.6$)	[20, 800]	0.15
Ausrüstung	Lognormal($\mu=5.0$, $\sigma=0.7$)	[20, 800]	0.05

Clipped prices are binned into tiers: **budget** (< 80), **mid** (80–180), **premium** (> 180).

Sales rank (product popularity). A Pareto-distributed popularity score S_i implements an 80/20 demand concentration:

$$S_i \sim \text{Pareto}(a = 2.5), \quad P_{\text{prod}}(i) = \frac{S_i}{\sum_j S_j}.$$

SCD Type 2 reclassifications. On May 1 and October 1 each year, 8–12% of SKUs are randomly selected for a category or price-tier change. Affected items receive a price adjustment drawn uniformly:

$$\Delta P \sim \begin{cases} \mathcal{U}(+20\%, +30\%) & (\text{increase}), \\ \mathcal{U}(-20\%, -15\%) & (\text{decrease}). \end{cases}$$

III. Seasonality & trend

Expected daily volume is scaled by a composite multiplier over the full timeline:

$$M_{\text{total}}(t) = M_{\text{trend}}(t) \times M_{\text{week}}(t) \times M_{\text{month}}(t) \times M_{\text{event}}(t).$$

Linear trend. Models 20% organic growth over two years:

$$M_{\text{trend}}(t) = 1.0 + \frac{0.20}{T} t.$$

Weekly seasonality. Encodes intra-week demand variation (weekend spike):

$$M_{\text{week}}(t) \in \{0.85, 0.90, 0.90, 0.95, 1.00, 1.20, 1.30\} \quad (\text{Tue} \rightarrow \text{Sun}).$$

Payday effect. End-of-month spending lift:

$$M_{\text{month}}(t) = \begin{cases} 1.15 & \text{if day-of-month} > 25, \\ 1.00 & \text{otherwise.} \end{cases}$$

Promotional events (step-function multipliers).

$$M_{\text{event}}(t) = \begin{cases} 1.5 & \text{Jul 15–30 (Summer Sale),} \\ 3.0 & \text{Nov 20–27 (Black Week),} \\ 1.8 & \text{Dec 1–15 (Christmas Rush),} \\ 0.2 & \text{Dec 24–26 (Christmas Slump),} \\ 1.0 & \text{otherwise.} \end{cases}$$

IV. Order generation (non-homogeneous Poisson process)

For each shop $k \in \{\text{DE}, \text{AT}, \text{CH}\}$ with baseline rates $\lambda_{\text{base}} = \{50, 15, 10\}$, the time-varying intensity is

$$\lambda_k(t) = \lambda_{\text{base},k} \times M_{\text{total}}(t).$$

The realised order count on day t is sampled as

$$N_{\text{orders}}(t, k) \sim \text{Poisson}(\lambda_k(t)), \quad P(X=n) = \frac{\lambda_k(t)^n e^{-\lambda_k(t)}}{n!}.$$

Because λ fluctuates with the seasonality product, the process is *non-homogeneous*, yielding realistic auto-correlated daily volumes rather than i.i.d. counts.

V. Basket composition & line items

Basket size. Items per order follow an empirical discrete distribution:

$$P(\text{size}=s) = \begin{cases} 0.50 & s = 1, \\ 0.30 & s = 2, \\ 0.15 & s = 3, \\ 0.05 & s = 4. \end{cases}$$

Discounts. Each line item independently receives a 10% discount with probability $p_{\text{disc}} = 0.05$:

$$P_{\text{paid}} = \begin{cases} 0.9 P_i & \text{w.p. } 0.05, \\ P_i & \text{w.p. } 0.95. \end{cases}$$

Shipping delay. Post-order shipping latency: $D_{\text{ship}} \sim \mathcal{U}\{1, 5\}$ days.

Returns. Whether a line item is returned is a Bernoulli trial with category-specific probability r :

$$R_i \sim \text{Bernoulli}(r), \quad r \in \{0.05, 0.15, 0.30\}.$$

If $R_i = 1$, the return processing offset is $D_{\text{return}} \sim \mathcal{U}\{7, 30\}$ days post-shipment.

VI. Summary of distributions

Component	Distribution	Parameters
Customer activity	Zipf	$a = 2.0$
Product price	Lognormal	$\mu \in \{4.4, 4.8, 5.0\}, \sigma \in \{0.5, 0.6, 0.7\}$
Product popularity	Pareto	$a = 2.5$
Daily orders	Poisson	$\lambda_k(t)$ (time-varying)
Basket size	Empirical discrete	$\{1, 2, 3, 4\}$
Discount trigger	Bernoulli	$p = 0.05$
Shipping delay	$\mathcal{U}\{1, 5\}$	discrete uniform
Return occurrence	Bernoulli	$r \in \{0.05, 0.15, 0.30\}$
Return delay	$\mathcal{U}\{7, 30\}$	discrete uniform